

# SoverGrid CLI

The one-click, un-censorable decentralized AWS.

 [Read the Official Developer Documentation](#)

SoverGrid is an open-source command-line tool that deploys your applications to decentralized networks (Akash, Filecoin, Spheron, Golem) with a single command. No rewriting code. No centralized gatekeepers. No surprise bills.

## Why SoverGrid?

Traditional cloud platforms (AWS, Railway, Vercel) can shut down your server, spike your bill, or censor your deployment at any time. SoverGrid routes your code to decentralized infrastructure where no single company controls the off switch.

**For Web2 Developers:** You don't need to learn blockchain. Write your normal Python or Node.js app, and SoverGrid handles the rest. It auto-generates Dockerfiles, calculates costs, and deploys to the cheapest available decentralized network.

**For Web3 Builders:** Pay for deployments with USDC or the \$SVR utility token. Every transaction is transparent, with fees automatically split via a smart contract.

## Current Status (Beta Phase 1 - Live Demo)

SoverGrid is currently in **Beta Phase 1**, designed specifically as a working proof-of-concept for investors and early developers.

**What is currently working 100% (The Financial Plumbing):**

- **Web3 Payment Routing:** When a developer runs `sovergrid deploy`, the CLI securely connects to the blockchain, signs a transaction with their private key, and charges them testnet USDC.
- **Smart Contract Revenue Split:** The 80/20 revenue split (where 20% goes to the SoverGrid Treasury and 80% to the compute provider) is executed flawlessly on-chain.
- **Orchestration Logic:** The CLI properly parses the `sovergrid.yaml` file, calculates costs, simulates latency, and checks for budget limits.

**What is mocked (Pending Phase 4 Funding):**

- **Decentralized Network Integration:** In this beta phase, the actual connection to Akash, Spheron, and Golem is mocked. The backend pretends to deploy the code and returns a simulated DNS link.
- **Next Steps:** With seed funding, Phase 4 will replace these mock endpoints with the real APIs for Akash, Filecoin, and Bittensor, turning the working financial engine into a fully operational decentralized cloud.

## Quick Start

### 1. Installation

SoverGrid CLI is cross-platform. Ensure you have Python 3.8+ installed.

### Mac / Linux:

```
pip3 install sovergrid
```

### Windows:

```
pip install sovergrid
```

## 2. Developer Authentication

Before deploying, you need to create a developer account. This allows you to manage deployments and (in the future) fund your wallet directly.

```
# Register a new account
sovergrid register
# (You will be prompted for an email and password)

# Log in to an existing account
sovergrid login
```

## 3. Get Testnet Funds (Beta Only)

Since the CLI requires testnet USDC to deploy, you can instantly fund your wallet using the built-in faucet:

```
sovergrid faucet
```

## 4. Deploy Your App

```
# Initialize your project
cd my-web-app
sovergrid init

# Deploy to the decentralized cloud
sovergrid deploy
```

## Commands

### Full Stack

| Command            | Description                                                      |
|--------------------|------------------------------------------------------------------|
| sovergrid register | Create a new SoverGrid developer account                         |
| sovergrid login    | Authenticate your CLI                                            |
| sovergrid logout   | Log out of the CLI                                               |
| sovergrid faucet   | Mint \$1,000 in free testnet USDC to test the CLI                |
| sovergrid init     | Scaffold a new project (generates sovergrid.yaml and Dockerfile) |
| sovergrid deploy   | Deploy your app to the decentralized network (compute + storage) |

|                               |                                              |
|-------------------------------|----------------------------------------------|
| <code>sovergrid status</code> | Check the status of your active deployment   |
| <code>sovergrid info</code>   | Display SoverGrid version and current config |

## Standalone Services

Each service works **completely independently**. You do not need to use all of them.

| Command                       | Service     | Description                                                                |
|-------------------------------|-------------|----------------------------------------------------------------------------|
| <code>sovergrid train</code>  | ML Training | Train AI models on decentralized GPUs (Bittensor, Gensyn, io.net)          |
| <code>sovergrid store</code>  | Storage     | Pin files to decentralized storage (Filecoin, Arweave, IPFS)               |
| <code>sovergrid db</code>     | Database    | Provision a decentralized database (Kwil SQL, Polybase NoSQL)              |
| <code>sovergrid cdn</code>    | CDN         | Distribute content via decentralized CDN (Fleek, Saturn)                   |
| <code>sovergrid secure</code> | Security    | Integrate Web3 privacy, encryption, and KMS (Lit Protocol, Secret Network) |

## Green Compute

SoverGrid supports eco-friendly deployments. By adding `green: true` to your `sovergrid.yaml`, your code will only be routed to providers that are powered by renewable energy. This feature ensures that your infrastructure has a lower carbon footprint, and it will expand as more green-certified DePIN nodes join the network.

## Configuration

SoverGrid uses a `sovergrid.yaml` file in your project root:

```
app:
  name: "my-web-app"

compute:
  provider: "akash"
  region: "us-west"
  resources:
    cpu: 1
    memory: "512Mi"

storage:
  provider: "filecoin"

security:
  provider: "lit-protocol"
  features:
    - "decentralized_kms"
    - "ddos_protection"

build:
  port: 8080
```

```
payment:
  token: "USDC"
  max_budget: 5.00
```

## Cost Breakdown

Every deployment fee is transparently split:

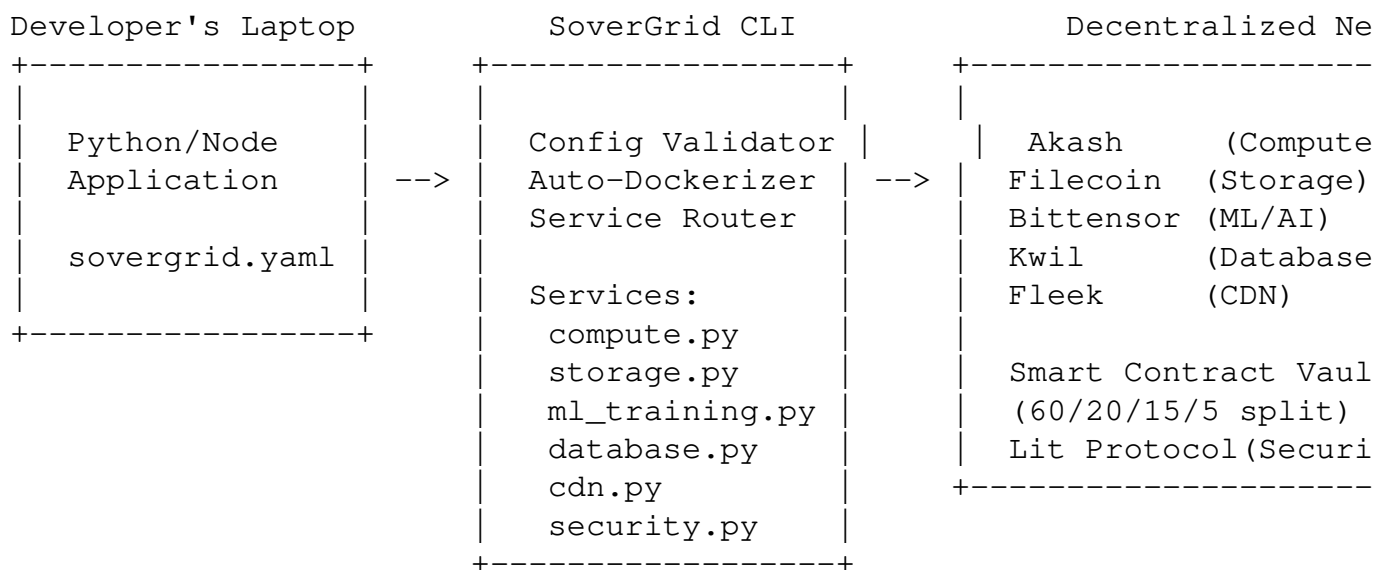
| Split              | Percentage | Purpose                                           |
|--------------------|------------|---------------------------------------------------|
| Compute Provider   | 60%        | Pays Akash/Spheron/Golem for server resources     |
| SoverGrid Treasury | 20%        | Funds development, legal, and marketing           |
| SVR Stakers        | 15%        | Rewards token holders who secure the network      |
| Auto-Liquidity     | 5%         | Automatically deepens the SVR/USDC liquidity pool |

## Supported Stacks

SoverGrid auto-detects your project and generates optimized Dockerfiles:

| Stack             | Detection                           |
|-------------------|-------------------------------------|
| Python            | requirements.txt                    |
| Python (FastAPI)  | requirements.txt containing fastapi |
| Node.js           | package.json                        |
| Node.js (Next.js) | package.json containing next        |

## Architecture



## How Deployment Works (Behind the Scenes)

When you run `sovergrid deploy`, here is what happens to your code:

1. **Zip & Ship:** The CLI securely zips your local project and sends it to the SoverGrid

Backend. It does *not* push your code to GitHub.

2. **Auto-Dockerizer:** The backend detects your framework (e.g., FastAPI, Next.js), auto-generates a `Dockerfile`, and builds your container image.
3. **Decentralized Registry:** Your compiled container image is pushed to the SoverGrid Container Registry (backed by decentralized storage like Filecoin). We use version tagging (`v1`, `v2`), meaning your old code is never deleted, and you can instantly rollback.
4. **Network Handoff:** The backend instructs the decentralized compute network (e.g., Akash) to pull your image from the registry and spin up your server.

This means you get the simplicity of Vercel/Heroku, with the censorship-resistance of Web3.

Each service is an **independent Python module**. They share a common `BaseService` abstract class but have zero dependencies on each other. A bug in `ml_training.py` cannot break `compute.py`.

## Examples

The `examples/` folder contains ready-to-use configs for every use case:

| File                               | Use Case                                     | Command                                                        |
|------------------------------------|----------------------------------------------|----------------------------------------------------------------|
| <code>sovergrid.yaml</code>        | All services combined (default)              | <code>sovergrid deploy</code>                                  |
| <code>compute-only.yaml</code>     | Just deploy an app to decentralized compute  | <code>sovergrid deploy -c examples/compute-only.yaml</code>    |
| <code>storage-only.yaml</code>     | Just pin files to Filecoin/IPFS              | <code>sovergrid store -c examples/storage-only.yaml</code>     |
| <code>ml-training-only.yaml</code> | Just train an AI model on decentralized GPUs | <code>sovergrid train -c examples/ml-training-only.yaml</code> |
| <code>database-only.yaml</code>    | Just provision a decentralized database      | <code>sovergrid db -c examples/database-only.yaml</code>       |
| <code>cdn-only.yaml</code>         | Just push content to a decentralized CDN     | <code>sovergrid cdn -c examples/cdn-only.yaml</code>           |
| <code>full-stack.yaml</code>       | Complete infrastructure (all 5 services)     | All commands                                                   |

## License

MIT